

$$2. \quad \overline{AB'C} + \overline{A'BC} + \overline{ABC'} + \overline{A'B'C} + \overline{AB'C'}$$

$$= \overline{B'C} (\overline{A+A'}) + \overline{AC'} (\overline{B+B'}) + \overline{A'BC}$$

$$= \overline{B'C} + \overline{AC'} + \overline{A'BC}$$

$$= \overline{C} (\overline{B'+A'B}) + \overline{AC'}$$

$$\begin{array}{c|c|c} A & B & \\ \hline 0 & 0 & \\ 0 & 1 & \\ 1 & 0 & \\ 1 & 1 & \end{array}$$

A	B	C	$\overline{B'C}$	$\overline{AC'}$	$\overline{A'BC}$	y
1	0	0	0	1	0	1
1	0	1	1	0	0	1
1	1	0	0	1	0	1
1	1	1	0	0	0	0
0	0	0	0	0	0	0
0	1	0	0	0	0	0
0	0	1	1	0	0	1
0	1	1	0	0	1	1

$$y = \text{Not} (\overline{ABC} + \overline{A'B'C'} + \overline{A'BC'})$$

$$= \overline{ABC} \times \overline{A'B'C'} \times \overline{A'BC'}$$

$$= (\overline{A'+B'+C'}) (\overline{A+B+C}) (\overline{A+B'+C})$$

$$= (\overline{A'B+A'C+B'A+B'C+C'A+BC'}) (\overline{A+B'+C})$$

$$= \overline{AB'C} + \overline{AB'} + \overline{AC'} + \overline{ABC'} +$$

$$= \text{Not} (\overline{ABC} + \overline{A'C'})$$

$$= (\overline{A'+B'+C'}) (\overline{A+C}) = \overline{A'C + AB' + B'C + AC'}$$

Consensus

$$= \overline{A'C} + \overline{AB'} + \overline{AC'}$$

$$= \overline{A \times C} + \overline{AB'}$$

$$\boxed{y = (\overline{A \times C}) \text{ or } (\overline{A \text{ and not } B})}$$